

Key safety points for test engineers

1. Keep the unauthorised and unqualified persons away from the test area
2. Arrange the test bench and test procedure in orderly manner
3. Do not modify the test procedures without any knowledge
4. Make sure that custom test fixtures carry proper ratings and test equipment are correctly rated
5. Do not touch the DUT while the test is running
6. In case of a problem, first turn off/stop the test
7. Discharge the DUT completely in the DC test

age cannot charge a capacitive load, as reactive current flows instantly and remains consistent regardless of the time the voltage is applied. DC voltage will charge the capacitance of the DUT so you may see an inrush of current upon initial application of the voltage, but gradually the readings will dissipate as the DUT becomes charged. Normally, the relationship between AC and DC HIPOT tests is $AC=1.414DC$. For example, if you perform AC HIPOT test at a test voltage of 2 kV, then DC HIPOT test should be performed at 2.83 kV," explains Ms Wang.

Expressing her views on the test methodology, Ms Gupta believes that the test method also depends on the type of the DUT. "For instance, when a low-capacitance equipment like electrical cabling or switchgear is tested, the time-dependent capacitive leakage and absorption leakage current become insignificant and decrease to zero almost instantly. This provides a perfect condition for accurate spot reading. On the other hand, when high-capacitance equipment such as long runs of cables, large motors or generators are to be tested, the time-dependent current will last hours. This current will cause the meter reading to change constantly, making it impossible to obtain an accurate steady reading. This condition can be overcome by using a test that establishes a trend between



Withstand voltage tester

readings, such as the step voltages or dielectric-absorption test. These tests do not depend on a single reading but on a collection of relative readings."

New features in HIPOT testers

To serve the rising demands for features and, most importantly, safety, there have been a lot of developments in the design of HIPOT testers and features. Most companies are promoting their all-in-one kinds of equipment that can perform dielectric breakdown, dielectric-withstand and insulation resistance tests from a single box. Many equipment are designed to perform both AC as well as DC HIPOT test.

Handheld portable testers are now the need of every industry. The need for insulation testing is increasing in the field and traditional testers are large and bulky. "Nowadays, battery-

operated insulation testers are there to work in the field. Good-quality probes are available in the market to do the test efficiently," informs Manwani.

Some T&M manufacturers have launched a thoughtful combination of insulation tester and digital multimeter in a lightweight tool that fits in your palm. This has significantly helped electricians and technicians on the field in troubleshooting and preventive maintenance, addressing their basic multimeter needs, and gives them an insulation tester that generates up to 1 kV.

"Humidity is a major factor affecting insulation measurements in India. Nowadays, insulation testers are designed to be humidity-proof," shares Thirumalaisami R., head of T&M, Yokogawa, India. "During the DC HIPOT test, the DUT is fully charged to provide accurate leakage current. Normally, to discharge the testers, the

TABLE I
Effects of Electric Current on the Human Body

Current	Reaction
1 mA	Perception level. Just a faint tingle
5 mA	Slight shock felt; not painful but disturbing Average individual can let go. However, strong involuntary reactions to shocks in this range can lead to injuries
6-25 mA (women)	Painful shock, muscular control is lost
9-30 mA (men)	This is called the freezing current or 'let-go' range
50-150 mA	Extreme pain, respiratory arrest, severe muscular contractions that an individual cannot let go. Death is possible
1000-4300 mA	Ventricular fibrillation (rhythmic pumping action of the heart ceases), muscular contraction and nerve damage. Death is most likely
10,000 mA	Cardiac arrest, severe burns and probable death

Source: W.B. Kouwenhoven, 'Human Safety and Electric Shock,' *Electrical Safety Practices*, Monograph, T12, Instrument Society of America, p. 93